

STAT 516: Homework 7

Instructions: Please submit your homework as a single PDF on Gradescope. After uploading, please make sure each page is assigned to the correct question before submitting. Show your work clearly.

1. **(15 points)** Let X and Y be continuous random variables with joint PDF

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{2}e^{-y}, & -y < x < y, \ y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate

$$P(X < 1 \mid Y = 3).$$

2. **(15 points)** Let X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 2x, & 0 < x < 1, \ 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate

$$P\left(X + Y \leq 1 \mid X \leq \frac{1}{2}\right).$$

3. **(15 points)** Let X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 24xy, & 0 < x < 1, \ 0 < y < 1 - x, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate

$$P\left(Y < X \mid X = \frac{1}{3}\right).$$

4. **(13 points)** A fair coin is tossed. If a head occurs, one fair die is rolled. If a tail occurs, two fair dice are rolled. Let Y be the total number of points showing on the die or dice. Calculate $E[Y]$.

5. **(15 points)** Let X and Y denote the values of two stocks at the end of a five-year period. Suppose X is uniformly distributed on the interval $(0, 12)$. Given $X = x$, suppose Y is uniformly distributed on the interval $(0, x)$.

Calculate

$$\text{Cov}(X, Y).$$

6. **(15 points)** Let X be the loss amount for a policyholder. Suppose X is uniformly distributed on the interval $[0, b]$.

The policy has a deductible of 180, which means the policyholder is responsible for paying the first 180 of each loss. Therefore, the unreimbursed portion of a loss is

$$\min(X, 180).$$

If

$$E[\min(X, 180)] = 144,$$

calculate b .

7. **(12 points)** Let

$$E[Y \mid X = x] = 2x, \quad \text{Var}(Y \mid X = x) = 4x^2,$$

and suppose X has a uniform distribution on the interval $(0, 1)$.

Calculate $\text{Var}(Y)$.

1. (15 points) Let X and Y be continuous random variables with joint PDF

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{2}e^{-y}, & -y < x < y, y > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate

$$P(X < 1 \mid Y = 3).$$

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} \textcircled{1} \rightarrow f_{X|Y}(x|3) = \frac{f_{X,Y}(x,3)}{f_Y(3)} \textcircled{1}$$

$$\textcircled{1} f_Y(y) = \int_{-y}^y \underbrace{\frac{1}{2}e^{-y}}_{\text{constant}} dx = \frac{1}{2}e^{-y} \int_{-y}^y [x]_{-y}^y = \frac{1}{2}e^{-y} (\underbrace{y}_{12} - \underbrace{(-y)}_{2y}) = ye^{-y}$$

$$f_Y(y) = ye^{-y}, y > 0$$

$$f_Y(3) = 3e^{-3}$$

$$f_{X|Y}(x|3) = \frac{f_{X,Y}(x,3)}{f_Y(3)} \textcircled{1} = \frac{\cancel{1/2}e^{-3}}{\cancel{3}e^{-3}} = \frac{1}{6}, -3 < x < 3$$

Conditional on $X|Y=3 \sim \text{Uniform}(-3, 3)$

$$P(X < 1 \mid Y=3) = \int_{-3}^1 \frac{1}{6} dx = \frac{1}{6} [x]_{-3}^1 = \frac{1}{6} (1 - (-3)) = \frac{4}{6}$$

$$P(X < 1 \mid Y=3) = 2/3$$

2. (15 points) Let X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 2x, & 0 < x < 1, 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate

$$P\left(X + Y \leq 1 \mid X \leq \frac{1}{2}\right).$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$\textcircled{1} P(X \leq 1/2) = \int_0^{1/2} \int_0^1 2x dy dx = \int_0^{1/2} 2x[y]_0^1 dx = \int_0^{1/2} 2x dx = [x^2]_0^{1/2}$$

$$P(X \leq 1/2) = 1/4$$

$$\textcircled{2} A \cap B = \{x+y \leq 1, x \leq 1/2\} \rightarrow \{0 \leq x \leq 1/2, 0 \leq y \leq 1-x\}$$

↓
 $y \leq 1-x$

$$\begin{aligned} P(x+y \leq 1, x \leq 1/2) &= \int_0^{1/2} \int_0^{1-x} 2x dy dx = \int_0^{1/2} 2x(1-x) dx = \int_0^{1/2} 2x - 2x^2 dx \\ &= \left[x^2 - \frac{2}{3}x^3\right]_0^{1/2} = \left(\frac{1}{2}\right)^2 - \frac{2}{3}\left(\frac{1}{2}\right)^3 = \frac{1}{4} - \frac{2}{3}\left(\frac{1}{8}\right) = \frac{1}{4} - \frac{1}{12} \end{aligned}$$

$$P(x+y \leq 1, x \leq 1/2) = \frac{1}{6}$$

$$P(x+y \leq 1 \mid x \leq 1/2) = \frac{P(x+y \leq 1, x \leq 1/2)}{P(x \leq 1/2)} = \frac{1/6}{1/4}$$

$$P(x+y \leq 1 \mid x \leq 1/2) = \frac{2}{3}$$

3. (15 points) Let X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 24xy, & 0 < x < 1, 0 < y < 1-x, \\ 0, & \text{otherwise.} \end{cases}$$

Calculate

$$P\left(Y < X \mid X = \frac{1}{3}\right).$$

$$f_X(x) = \int_0^{1-x} 24xy \, dy = 24x \left[\frac{y^2}{2} \right]_0^{1-x} = 12x(1-x)^2, \quad 0 < x < 1$$

$$f_{X|Y}(y|x) = \frac{f_{X,Y}(x,y)}{f_Y(x)} = \frac{24xy}{12x(1-x)^2} = \frac{2y}{(1-x)^2}, \quad 0 < y < 1-x$$

$$f_{X|Y}(y|\frac{1}{3}) = \frac{2y}{(1-\frac{1}{3})^2} = \frac{2y}{4/9} = \frac{9}{2}y, \quad 0 < y < \frac{2}{3}$$

$\downarrow$
 $1 - 1/3$

$$P(Y < X \mid X = \frac{1}{3}) = \int_0^{1/3} \frac{9}{2}y \, dy = \frac{9}{2} \left[\frac{y^2}{2} \right]_0^{1/3} = \frac{9}{4} \left(\frac{1}{9} \right)$$

$$P(Y < X \mid X = \frac{1}{3}) = \frac{1}{4}$$

4. (13 points) A fair coin is tossed. If a head occurs, one fair die is rolled. If a tail occurs, two fair dice are rolled. Let Y be the total number of points showing on the die or dice.

Calculate $E[Y]$. Number of points depend on coin result.

$$\begin{aligned} E[Y] &= E[E(Y \mid \text{coin result})] \\ &= P(H)E(Y|H) + P(T)E(Y|T) \\ &= \left(\frac{1}{2}\right)\left(\frac{7}{2}\right) + \left(\frac{1}{2}\right)(7) \\ &= \frac{7}{4} + \frac{14}{4} = \frac{21}{4} \end{aligned}$$

$$E[Y|H] = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = \frac{7}{2} \text{ one die rolled}$$

$$E[Y|T] = 2\left(\frac{7}{2}\right) = 7 \text{ two dice rolled}$$

$$E[Y] = 21/4 = 5.25$$

5. (15 points) Let X and Y denote the values of two stocks at the end of a five-year period. Suppose X is uniformly distributed on the interval $(0, 12)$. Given $X = x$, suppose Y is uniformly distributed on the interval $(0, x)$.

Calculate

$$\frac{a+b}{2}$$

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

① $E[X]$ and $E[X^2]$

$$X \sim \text{Uniform}(0, 12)$$

$$E[X] = \frac{0+12}{2} = 6 \quad E[X^2] = \frac{1}{12} \int_0^{12} x^2 dx = \frac{1}{12} \left[\frac{x^3}{3} \right]_0^{12} = 48$$

② $E[Y]$

$$Y|X=x \sim \text{Uniform}(0, x) \quad f_X(x) = \frac{1}{12}, \quad 0 < x < 12$$

$$E[Y|X=x] = \frac{x}{2} = \frac{0+x}{2}$$

Law of Total Expectation: $E[Y] = E[E[Y|X]] = E\left[\frac{X}{2}\right] = \frac{1}{2} E[X] = \frac{1}{2}(6)$
 $E[Y] = 3$

③ $E[XY]$

Conditional Expectation: $E[XY] = E[E[XY|X]] = X E[Y|X] = E\left[X\left(\frac{X}{2}\right)\right] = \frac{1}{2} E[X^2] = \frac{1}{2}(48)$
 $E[XY] = 24$ (fixed)

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 24 - (6)(3)$$

$$\text{Cov}(X, Y) = 6$$

6. (15 points) Let X be the loss amount for a policyholder. Suppose X is uniformly distributed on the interval $[0, b]$. $f_X(x) = \frac{1}{b-a}$, $0 \leq x \leq b$

The policy has a deductible of 180, which means the policyholder is responsible for paying the first 180 of each loss. Therefore, the unreimbursed portion of a loss is

$$\min(X, 180) = \begin{cases} x, & 0 \leq x \leq 180 \\ 180, & x > 180 \end{cases}$$

If

$$E[\min(X, 180)] = 144,$$

calculate b .

$$\text{If } b \leq 180, \min(x, 180) = x$$

$$E[\min(x, 180)] = \frac{b}{2} = 144$$

$$\text{⊖ } b = 288, \text{ but } 288 > 180 \times$$

Therefore, $b > 180$.

$$E[\min(x, 180)] = \int_0^{180} x \frac{1}{b} dx + \int_{180}^b 180 \frac{1}{b} dx = 144$$

$$= \frac{1}{b} \left[\frac{x^2}{2} \right]_0^{180} + \frac{180}{b} [x]_{180}^b = 144$$

$$= \frac{180^2}{2b} + \frac{180(b-180)}{b} = 144$$

$$= \frac{16200 + 180b - 32400}{b} = 144$$

$$= 180 - \frac{16200}{b} = 144$$

$$= 36 = \frac{16200}{b}$$

$$b = 450$$

7. (12 points) Let

$$E[Y \mid X = x] = 2x, \quad \text{Var}(Y \mid X = x) = 4x^2,$$

and suppose X has a uniform distribution on the interval $(0, 1)$.

Calculate $\text{Var}(Y)$. = $E[\text{Var}(Y|X)] + \text{Var}(E[Y|X])$ Law of Total Variance

$$\textcircled{1} E[\text{Var}(Y|X)] = E[4X^2] = 4E[X^2]$$

$X \sim \text{Uniform}(0, 1)$

$$E[X^2] = \int_0^1 x^2 dx = \left[\frac{x^3}{3} \right]_0^1 = \frac{1}{3}$$

$$E[\text{Var}(Y|X)] = 4\left(\frac{1}{3}\right) = \frac{4}{3}$$

$$\textcircled{2} \text{Var}(E[Y|X]) = \text{Var}(2X) = 4\text{Var}(X)$$

$X \sim \text{Uniform}(0, 1)$ ↓

$$\text{Var}(aX) = a^2 \text{Var}(X)$$

$$\text{Var}(X) = \frac{(1-0)^2}{12} = \frac{1}{12}$$

$$\text{Var}(E[Y|X]) = 4\left(\frac{1}{12}\right) = \frac{1}{3}$$

$$\text{Var}(Y) = E[\text{Var}(Y|X)] + \text{Var}(E[Y|X]) = \frac{1}{3} + \frac{4}{3}$$

$$\text{Var}(Y) = \frac{5}{3}$$